Anand Parameter Extraction
04/11/2026

Solder joints in electronic assemblies experience cyclic thermal loading arising from power cycling and ambient temperature excursions. The resulting thermomechanical fatigue (TMF)—fatigue driven by the mismatch in coefficients of thermal expansion between dissimilar materials—governs the service life of the assembly. Accurate finite-element simulation of this damage mechanism requires a constitutive model that captures the rate-dependent (*i.e.*, time-dependent, or viscoplastic) and temperature-dependent inelastic behavior of the solder.

The Anand model, introduced by *Loveish Anand* in 1985 and subsequently extended by *Brown, Kim*, and *Anand* (1989), is a single internal-state-variable viscoplastic theory that simultaneously accounts for isotropic strain hardening and recovery (softening). It has become the de facto standard in the microelectronics reliability community because:

1. It requires no explicit yield surface, thereby eliminating the need to distinguish between elastic and plastic regimes.
2. A single scalar deformation resistance s serves as the only internal state variable, keeping implementation tractable.
3. Its nine material constants can be determined from conventional isothermal mechanical tests.

Governing Equations of the Anand Model

The flow rule specifies the rate of inelastic (plastic) strain $\dot{\varepsilon}_p$ as a function of the applied stress σ, the absolute temperature T (in Kelvin), and the internal state variable s:

$$\dot{\varepsilon}_p = A \exp\left(-\frac{Q}{RT}\right) \left[\sinh\left(\xi \frac{\sigma}{s}\right)\right]^{1/m}$$

Parameter Definitions

Symbol	Unit	Physical meaning
A	s^{-1}	Pre-exponential factor (frequency factor)
Q	$J\ mol^{-1}$	Activation energy for plastic flow
R	$8.314\, J\ mol^{-1} K^{-1}$	Universal gas constant
ξ	—	Multiplier of stress level (stress sensitivity coefficient)
m	—	Strain rate sensitivity exponent ($0 < m \leq 1$)
s	MPa	Deformation resistance (internal state variable)
T	K	Absolute temperature

The $\sinh(\cdot)^{1/m}$ term provides a smooth transition from power-law creep (dominant at low stresses, $\xi\sigma/s \ll 1$, where $\sinh x \approx x$) to exponential creep (dominant at high stresses, $\xi\sigma/s \gg 1$, where $\sinh x \approx \frac{1}{2}e^x$). The ratio σ/s is a normalized stress: it measures applied stress relative to the current resistance of the microstructure.

© All rights reserved, Orange County Science Institute, 2026

The deformation resistance s evolves according to a hardening–dynamic recovery equation:

$$\dot{s} = \left[h_0 \left| 1 - \frac{s}{s^*} \right|^a \operatorname{sgn}\left(1 - \frac{s}{s^*}\right) \right] \dot{\varepsilon}_p$$

where s^* is the saturation value of s, itself a function of temperature and strain rate:

$$s^* = \hat{s} \left[\frac{\dot{\varepsilon}_p}{A} \exp\left(\frac{Q}{RT}\right) \right]^n$$

Additional parameter definitions:

Symbol	Unit	Physical meaning
h_0	MPa	Hardening/softening constant
a	—	Strain rate sensitivity of hardening ($a \geq 1$)
$\hat{s}$	MPa	Coefficient for saturation deformation resistance
n	—	Strain rate sensitivity of saturation resistance
s_0	MPa	Initial value of s at the start of deformation

The function $\operatorname{sgn}(1 - s/s^*)$ ensures that s hardens when $s < s^*$ and recovers (softens) when $s > s^*$. At steady state, $s = s^*$ and $\dot{s} = 0$.

The Nine Anand Constants

The equations contain exactly nine material constants:

$$\mathcal{P} = \{s_0,\ Q/R,\ A,\ \xi,\ m,\ h_0,\ \hat{s},\ n,\ a\}$$

All nine must be identified from experimental data.

Steady-State (Creep) Simplification

At steady state, $\dot{s} = 0, s = s^*$, and the flow rule reduces to a pure creep equation. Inverting the first equation at steady state yields the stress–strain-rate–temperature surface:

$$\sigma_{ss} = \frac{s^*}{\xi} \sinh^{-1}\left[\left(\frac{\dot{\varepsilon}_p}{A}\right)^m \exp\left(\frac{mQ}{RT}\right) \right]$$

This expression is central to the parameter extraction procedure because steady-state data are the most reliably obtained experimental quantities.

<u>Experimental Measurements</u>

Parameter extraction relies on two complementary categories of isothermal mechanical test performed on bulk solder specimens or representative coupons:

© All rights reserved, Orange County Science Institute, 2026

Type A — Constant Strain Rate (CSR) Tests

The specimen is deformed at a prescribed constant engineering strain rate $\dot{\varepsilon}$ while the stress σ is recorded as a function of strain ε. Each test produces a stress–strain curve at a given ($\dot{\varepsilon}$, T) pair. The peak (or plateau) stress at large strain is taken as the steady-state flow stress σ_{ss}.

Type B — Stress Relaxation Tests

The specimen is loaded to a target strain and then held at fixed total strain while the stress decays over time. The decaying stress records the inelastic strain rate indirectly via:

$$\dot{\varepsilon}_p = -\frac{\dot{\sigma}}{E(T)}$$

where $E(T)$ is the temperature-dependent Young's modulus (a separately measured elastic property, not an Anand parameter).

To adequately sample the constitutive surface, tests should span:

- Temperature range: From $-40\,^{\circ}\mathrm{C}$ to $150\,^{\circ}\mathrm{C}$ (or $0.4T_m$ to $0.9T_m$, where T_m is the alloy melting point in Kelvin), typically at 4–6 isothermal levels.
- Strain rate range: At least three decades, *e.g.*, 10^{-5}, 10^{-4}, 10^{-3}, $10^{-2}\ s^{-1}$.

This produces a minimum of ~ 12 independent (σ_{ss}, $\dot{\varepsilon}$, T) data triplets for regression.

<u>Parameter Extraction Procedure</u>

The extraction proceeds in two sequential stages:

- Stage 1 (Steady-State Analysis): Determine A, Q/R, ξ, m from the steady-state flow stresses.
- Stage 2 (Transient Analysis): Determine h_0, $\hat{s}$, n, a, and s_0 from the transient portions of the stress–strain curves or relaxation traces.

Stage 1: Steady-State Parameter Identification
Taking the natural logarithm of equation (1) at steady state ($s = s^*$):

$$\ln \dot{\varepsilon}_p = \ln A - \frac{Q}{RT} + \frac{1}{m} \ln\left[\sinh\left(\xi \frac{\sigma_{ss}}{s}\right)\right]$$

Because $s^* = s^*(\dot{\varepsilon}_p, T)$ appears on both sides, a direct linear regression is not possible in general. However, a two-step linearization is commonly employed:

At fixed T, above equation can be written:

$$\ln \dot{\varepsilon}_p = C_T + \frac{1}{m} \ln[\sinh(\beta_T\, \sigma_{ss})]$$

© All rights reserved, Orange County Science Institute, 2026

where $C_T = \ln A - Q/(RT)$ and $\beta_T = \xi/s^(T)$ are temperature-specific constants. A plot of $\ln \dot{\varepsilon}_p$ versus $\ln[\sinh(\beta_T\, \sigma_{ss})]$ should be linear with slope $1/m$. The value of β_T is found iteratively (or by scanning) to make this plot as linear as possible.

The intercept $C_T = \ln A - Q/(RT)$ is linear in $1/T$. Fitting C_T vs. $1/T$ over all temperatures yields slope $-Q/R$ and intercept $\ln A$.

The four Stage-1 constants (A, Q/R, ξ, m) are refined simultaneously by minimizing the sum of squared residuals between the observed and model-predicted steady-state stresses:

$$\min_{A,\, Q/R,\, \xi,\, m} \sum_{i=1}^{N} \left[\sigma_{ss,i}^{\text{obs}} - \sigma_{ss,i}^{\text{model}}\right]^2$$

where $\sigma_{ss,i}^{\text{model}}$ is evaluated at ($\dot{\varepsilon}_i$, T_i) with $s^* = \hat{s}$ (treated as a free parameter in Stage 1 or estimated from the upper bound of σ_{ss}).

Stage 2: Transient Parameter Identification
With the Stage-1 constants fixed, equations are integrated numerically as an initial-value problem for each CSR or relaxation test. The Stage-2 constants (h_0, $\hat{s}$, n, a, s_0) are varied to minimise:

$$\min_{h_0,\, \hat{s},\, n,\, a,\, s_0} \sum_j \sum_k \left[\sigma_{jk}^{\text{obs}} - \sigma_{jk}^{\text{model}}\right]^2$$

where the outer sum is over tests j and the inner sum over strain points k within each test.

Numerical Integration of the Anand Equations
Given a constant plastic strain rate $\dot{\varepsilon}_p$ (approximately the imposed total strain rate for large inelastic strains), the coupled ODEs are:

$$\frac{ds}{d\varepsilon_p} = \frac{\dot{s}}{\dot{\varepsilon}_p} = h_0 \left|1 - \frac{s}{s^*}\right|^a \operatorname{sgn}\left(1 - \frac{s}{s^*}\right)$$

$$\sigma = \frac{s}{\xi}\, \sinh^{-1}\left[\left(\frac{\dot{\varepsilon}_p}{A}\right)^m \exp\left(\frac{mQ}{RT}\right)\right]$$

Equation above is a scalar ODE that can be solved with standard *Runge-Kutta* or LSODA methods.

<u>Solder Alloy Examples</u>

Alloy	s_0 (MPa)	Q/R (K)	A (s^{-1})	ξ	m	h_0 (MPa)	$\hat{s}$ (MPa)	n	a
63Sn/37Pb	12.41	9400	4.0×10^6	11.0	0.303	1723.6	73.81	0.07	1.28
SAC305	1.576	8900	3.25×10^6	3.33	0.182	3321.0	58.0	0.02	1.82
Pb–5Sn	15.09	10830	1.5×10^7	7.1	0.228	1787.5	60.18	0.143	1.34

© All rights reserved, Orange County Science Institute, 2026

Not all nine Anand constants are equally sensitive to changes in the predicted response:

- Q/R and A: Strongly constrained by the temperature dependence of steady-state flow stress.
- m: Controlled by the slope of $\log \dot{\varepsilon}$ vs. $\log[\sinh(\cdot)]$ at fixed T. Insufficient strain-rate range (less than two decades) leads to poor identifiability.
- h_0 and a: Control the initial slope and shape of the transient hardening region. Poorly identified if tests do not capture small-strain behavior ($\varepsilon_p < 1\%$).
- n: Controls the rate sensitivity of the saturation stress. Nearly uncorrelated with h_0 if the deformation is rapid.

Practitioners should perform a sensitivity analysis — *e.g.*, by computing the Jacobian of the model predictions with respect to each parameter — before finalizing a parameter set.

The three alloys span a wide range of mechanical character:

- 63Sn/37Pb (eutectic tin–lead): Moderate creep resistance, intermediate Q/R. This alloy is the historical benchmark and has the most extensive published parameter database.
- SAC305 (95.5Sn–3.0Ag–0.5Cu): Lead-free replacement alloy. Higher h_0 and lower m indicate stiffer initial hardening and lower strain-rate sensitivity, reflecting the strengthening effect of Ag_3Sn and Cu_6Sn_5 intermetallic precipitates.
- Pb–5Sn (high-lead): Highest Q/R, indicating a higher activation energy and hence better elevated-temperature creep resistance. This alloy is used in high-reliability applications (*e.g.*, aerospace, power electronics) where SAC or Sn–Pb eutectic would be insufficient.

After extracting parameters, validation should include:

1. Leave-one-out cross-validation: Fit on $N-1$ data points, predict the held-out point. Repeat for all points.
2. Independent test comparison: Compare predictions against tests not used in fitting (different specimen geometry or loading profile).
3. FEA benchmark: Embed the parameters in a finite-element code and simulate a well-documented solder joint fatigue experiment; compare predicted fatigue life to observation.

<u>Concluding Remarks</u>

The Anand viscoplastic model provides a thermodynamically consistent and computationally tractable description of solder behavior under cyclic thermomechanical loading. The nine material constants are extracted in two sequential stages: a steady-state regression (Stage 1) that determines the flow-rule parameters $\{A, Q/R, \xi, m\}$ from isothermal creep data, followed by a transient regression (Stage 2) that identifies the hardening-recovery parameters $\{h_0, \hat{s}, n, a, s_0\}$ from the full stress–strain curves. The R codes provided implement this workflow for three representative solder alloys—63Sn/37Pb, SAC305, and Pb–5Sn—and can be adapted to any alloy for which sufficient isothermal mechanical test data are available.

© All rights reserved, Orange County Science Institute, 2026